ModelIt!

Student Activity Pack

The Crime Scene in Every Rock

Reading Earth's History One Layer at a Time

Grade 7 | NGSS MS-ESS1-4 | Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What can you learn by looking at layers of rock in a cliff or canyon?

2 Why do you think fossils are found inside certain types of rocks?

3 If you found a fish fossil in a rock on top of a mountain, what would that tell you?

4 Draw what you think rock layers look like underground.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sedimentation Rate | Erosion Rate | Rock Layer Thickness | Fossil Preservation Quality

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Rapid Deposition

Sedimentation Rate at maximum (flood event)

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Heavy Erosion

Erosion Rate at maximum, normal sedimentation

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Balanced System

Moderate sedimentation and erosion

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Tectonic Activity — Earthquakes, folding, and faulting can tilt, bend, or break rock layers after they form

Sea Level Change — Rising and falling oceans change what type of sediment is deposited in an area

Volcanic Eruptions — Volcanic ash creates distinct marker layers that can be dated and traced across continents

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How can a single rock tell you what happened on Earth millions of years ago? Use evidence from your simulation to support your answers.

1. **Why is the fossil record incomplete — why don't we have fossils of every organism that ever lived?**

2. **How does finding marine fossils on mountaintops prove that Earth's surface has changed dramatically?**

3. **What's the difference between relative age (superposition) and absolute age (radiometric dating)?**

4. **Why are sedimentary rocks the best type for finding fossils?**

5. **How does erosion act like a thief, stealing pages from Earth's history book?**

8 STEM Challenge: Create a Rock Core Sample Model

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YOUR ENGINEERING MISSION

Build a physical model of a rock core sample showing at least 5 distinct geological periods, each with appropriate fossils and environmental evidence.

THE SCENARIO

A museum of natural history needs an educational core sample exhibit showing how geologists read Earth's history from rock layers. Your team will build the model and write the interpretive guide.

Engineering Design Thinking

1. HOW MANY GEOLOGICAL PERIODS WILL YOUR CORE SAMPLE REPRESENT?
2. WHAT MATERIALS WILL REPRESENT DIFFERENT TYPES OF SEDIMENT AND ROCK?
3. WHAT FOSSILS OR EVIDENCE WILL YOU INCLUDE IN EACH LAYER?
4. HOW WILL YOU SHOW AT LEAST ONE EROSION GAP (UNCONFORMITY) IN YOUR RECORD?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Geologists and Stratigraphers study rock formations to understand Earth's history, locate natural resources, and assess geological hazards. They earn $65,000–$115,000/year.